# USBSID-Pico software manual

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# Disclaimer

I do this stuff in my free time for my enjoyment. Since I like to share my joy in creating this with everyone I try my best to provide a working PCB and Firmware. I am in no way an electronics engineer and can give ***no guarantees*** that this stuff does not break or damage your hardware, computer, phone, or whatever you try to hook it up to. Be sure to take great care when inserting any real MOS SID chips into the board. While everything has been tested with real chips, this is in no way a guarantee that nothing could go wrong. Use of this board and firmware at your own risk! I am in no way responsible for your damaged hardware. That being said, have fun!

# Supported platforms

For configuring your board itself (not the software listed below), see the `USBSID-Configtool-manual` file for the GUI tool, or the `USBSID-Pico-firmware-manual` file for the command line `cfg_usbsid` tool.
*In active development*
**Linux:** Vice, JSidplay2, Denise, SidplayFp, RetroDebugger, SidBerry, USB Midi, WebUSB & ASID (in webbrowser) SID Play
**Windows:** Vice, JSidplay2, Denise, SidBerry, USB Midi, WebUSB & ASID (in webbrowser) SID Play
**Android:** USB Midi, WebUSB & ASID (in webbrowser) SID Play
**Amiga:** PlaySID library

# Drivers

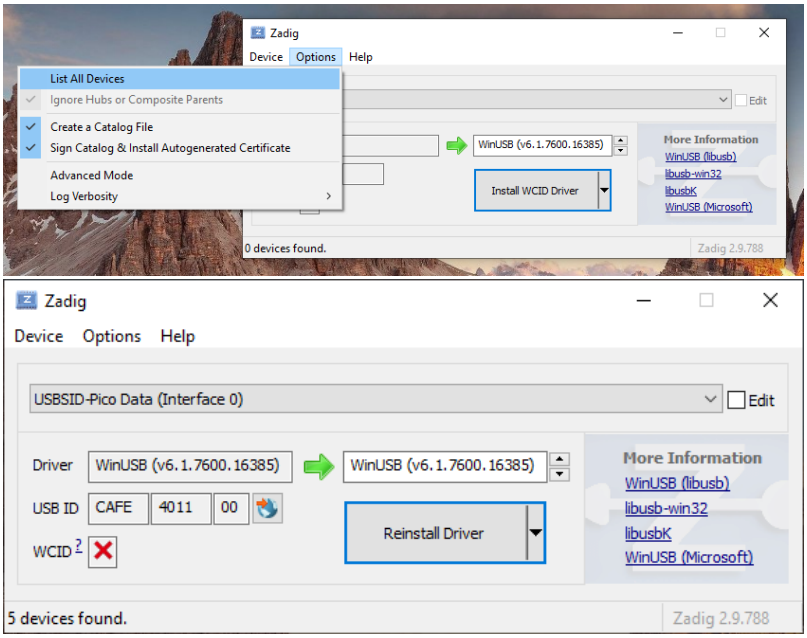
## Linux Udev rules

In the examples/udev directory you can find the udev rules that are used on Linux.
This purely an example file that you can use and change to your own needs. Steps required for this to work

```
# Check if you are in the plugdev group with the groups command
groups
# If not, add yourself to the plugdev group, then log out and in
sudo usermod -aG plugdev $USER
# Copy the udev ules file to the correct directory
sudo cp 69-usbsid-permissions.rules /etc/udev/rules.d
# Now reload the udev rules
udevadm control --reload-rules && udevadm trigger
# Not working? Try reloading the service
sudo systemctl restart udev
```

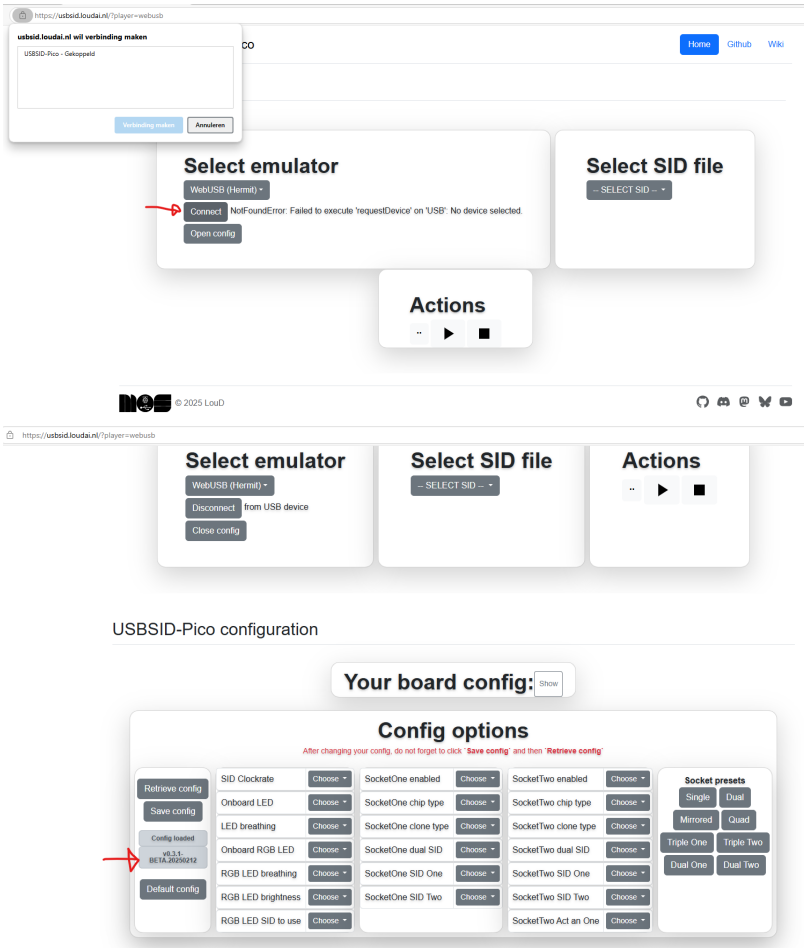
## Windows driver

Use Zadig to install the correct driver. Replace the driver for `USBSID-Pico Data` with WinUSB.




# Test and configure

After setting up your linux udev rules or Windows driver check, configure and test your board on the USBSID config tool website




USBSID-Pico configuration




*For more information on how to configure your board, please refer to the Firmware manual*

## USB connection: hubs and cables

Highly demanding cycle-exact SID tunes (digi/sample tunes) send thousands of small USB packets per second to the board. The board plays each packet on arrival and only buffers a few writes, USB delay is audible as crackles, clicks or short dropouts.
How the board is connected matters for these tunes:

- Connect the board directly to a USB port of your computer for the best result.
- When using a hub, use a USB 2.0 (480 Mbit/s) or newer hub. The board is a full speed (12 Mbit/s) device and the hub converts its traffic for the faster bus. A hub with one transaction translator per port (multi-TT) does this better than a single-TT hub.
- Do not chain hubs and avoid sharing the hub with busy devices such as audio interfaces, webcams or storage.
- Use a short, good quality USB data cable, avoid extension cables.
- Other software that talks to the board while a tune plays (for example a second player or tool polling the board) adds USB traffic and can cause the same artifacts.

Normal tunes are not affected by any of the above.

# Supported software

Each section covers the basic use of USBSID-Pico with the software under that section. For more in depth information please refer to the repective website and or github repo of the software.

## Deepsid

On the deepsid website you can select `WebUSB (Hermit)` or `ASID (MIDI)` for playing PSID files using a supported browser.

**WebUSB**: After selecting `WebUSB (Hermit)` you need to click the `CONNECT` button and select USBSID-Pico in the pop-up window. Due to the autoplay feature of Deepsid, you might have to press `STOP` a couple of times before the `CONNECT` button will work.

**ASID**: After selecting `ASID (MIDI)` you need to select USBSID-Pico in the dropdown box next to the line that says `MIDI port for ASID`. It might require refreshing the page a couple of times for the box to be filled with midi devices.
Supported browsers:

- **Firefox** supports ASID.
- **Chrome** supports both WebUSB and ASID.
- **Edge** supports both WebUSB and ASID.

# SidBerry

SidBerry supports PSID files only and can play SID files on USBSID-Pico via CDC (default) or on ASID supporting players via command line arguments.
General Linux usage:

```
# to play sidfile.sid on USBSID-Pico
usbsidberry path/to/sidfile.sid
# to detect and print available midi devices
usbsidberry -midi
# to play sidfile.sid via ASID on USBSID-Pico where `1` is the respective port of the
ASID device
usbsidberry -asid 1 path/to/sidfile.sid
```

Basic Windows usage:

```
# to play sidfile.sid on USBSID-Pico
usbsidberry.exe path/to/sidfile.sid
# to detect and print available midi devices
usbsidberry.exe -midi
# to play sidfile.sid via ASID on USBSID-Pico where `1` is the respective port of the
ASID device
usbsidberry.exe -asid 1 path/to/sidfile.sid
```

# Vice

This section covers the use of USBSID-Pico with Vice under Linux, Windows and MacOS for the GTK and SDL2 versions of `vsid` and `x64sc`

| | |
|---|---|
| **WARNING** | Switching on readmode and then switching it off again without restarting the PCB and Vice *will* result in distorted audio |

## Command-line arguments

Command-line arguments override any previously configured items in your `vicerc` config file.
All three OS support the same command-line arguments for both `vsid` and `x64sc`.
However, the sound device argument is OS specific and `vsid` supports a maximum of 3 SID chips
**Note** that if you define extra SID's and their addresses, these addresses need to point to the required address for the SID tune or Software you want to use. The Vice USBSID-Pico driver will automatically convert each write to the in firmware configured SID chip.
Below is an excerpt from the Vice manual with information about each command-line option

```
-usreadmode <0 or 1>
    Enable USBSID-Pico read mode. When enabled, this mode allows for reading from SID
registers that provides a means to configure an FPGASID that is seated on USBSID-Pico.
This mode is not suited for regular audio use as it disables cycle exact writing &
digiplay. Default is (SidUSBSIDReadMode=0) for off, (SidUSBSIDReadMode=1) for on.

-usaudiomode <0 or 1>
    Set USBSID-Pico PCB (v1.3) audio mode to Stereo or Mono. When enabled the audio of
SID socket 1 will be output over the left audio channel and the audio of SID socket 2
over the right channel of USBSID-Pico's audio out. Default is (SidUSBSIDAudioMode=0)
for Mono, (SidUSBSIDAudioMode=1) for Stereo.

-usdiffsize <number divisable by 8>
    Specifies the minimal difference in bytes to use between the head and tail of the
drivers ringbuffer, this allows for some audio quality tuning on slower systems. In
most cases the default value of 64 bytes (also the size of a USB packet) is the right
setting for best audio quality. (SidUSBSIDDiffSize=32-256).

-usbuffsize <number divisable by 8>
    Specifies the size in bytes to use as ringbuffer to store temporary data when
sending SID writes to USBSID-Pico, this allows for some audio quality tuning on slower
systems. In most cases the default value of 8192 bytes is the right setting for best
audio quality. (SidUSBSIDBufferSize=512-16384).
```

## x64sc GTK

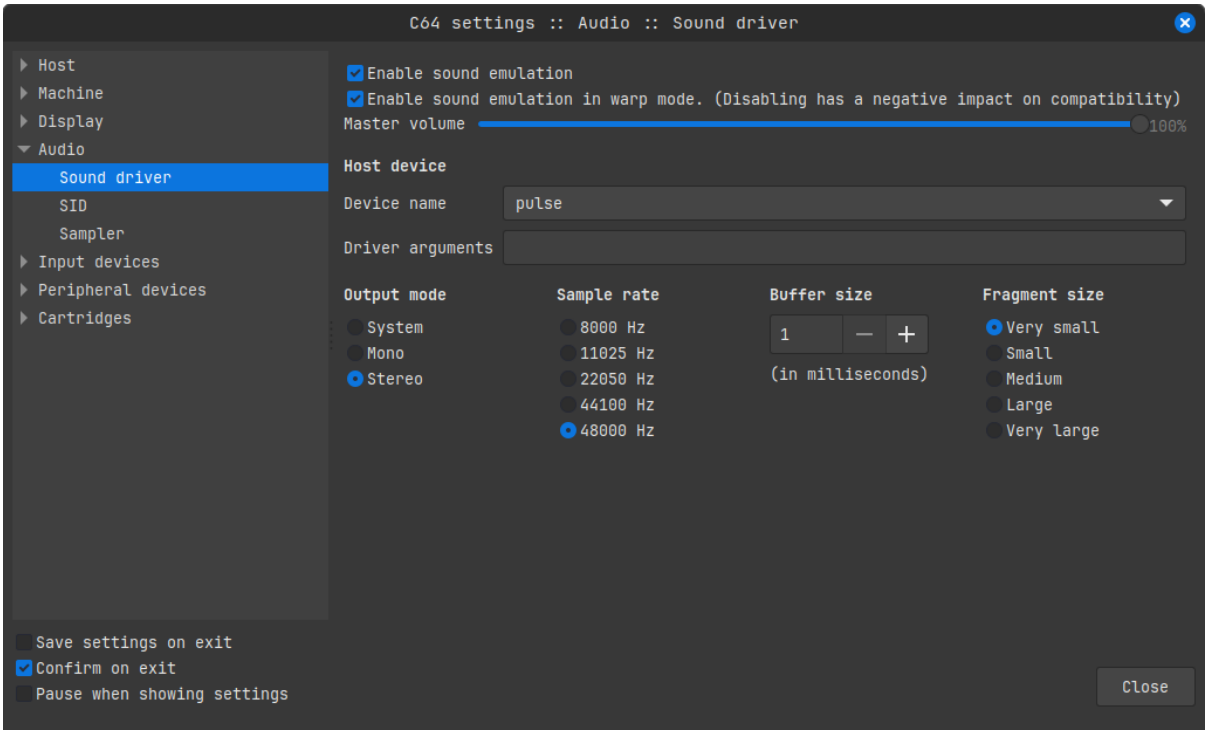
ALT+O or click Preferences/Settings to open the settings menu.

**Sound configuration**

The sound configuration will require tuning for the computer you use it on.

In the screenshot below you can see that for this system I use the `pulse` audio driver and have the `Buffer size` set to 1 millisecond and `Fragment size` set to very small.

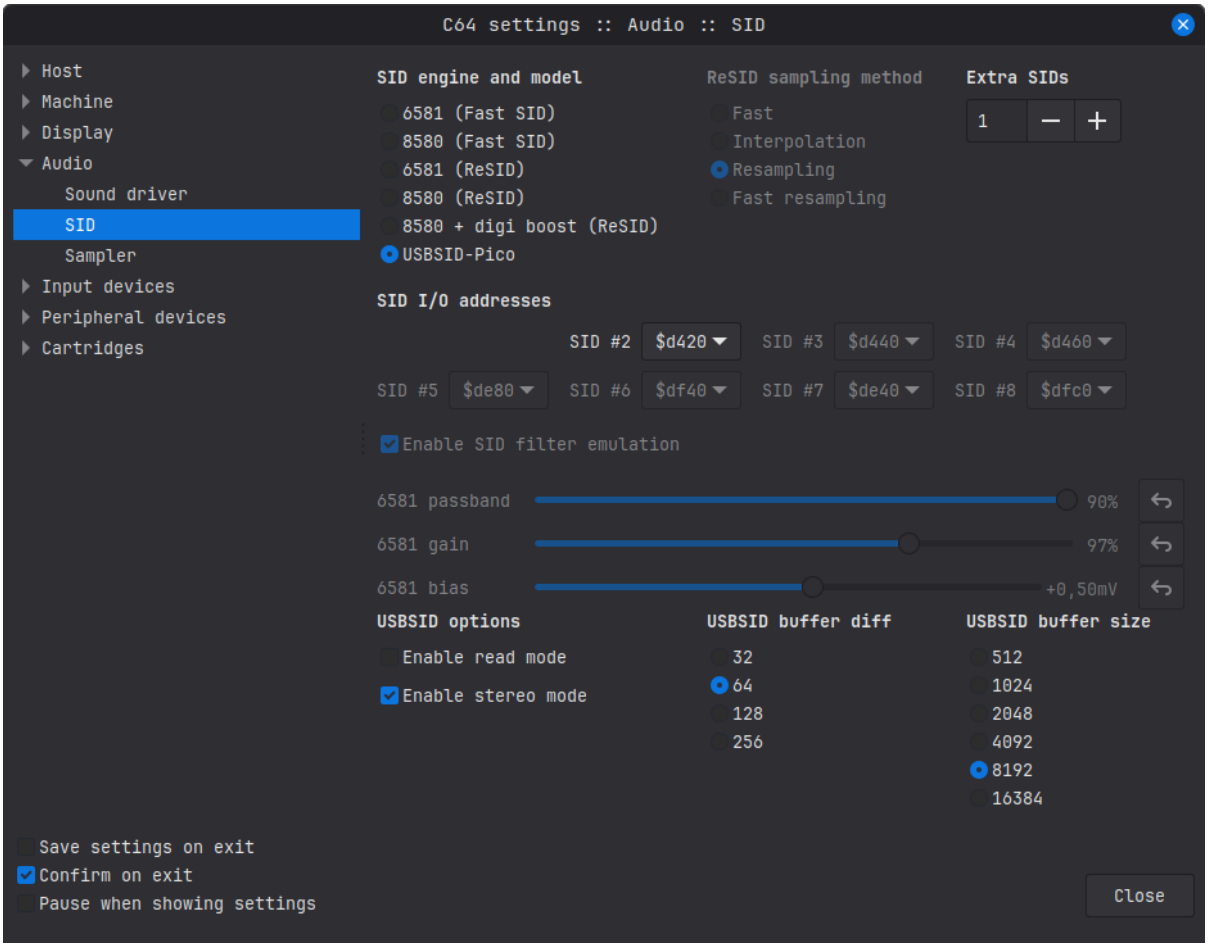
These 3 settings can make or break the audio quality.




**SID configuration**

For any audio to come out of the board you *need* to select USBSID-Pico as SID engine.

If you want to use more the one SID then you need to select the amount of `Extra SIDs` and set their I/O addresses.


C64 settings :: Audio :: SID
Host
Machine
Display
Audio
Sound driver
SID
Sampler
Input devices
Peripheral devices
Cartridges
SID engine and model
6581 (Fast SID)
8580 (Fast SID)
6581 (ReSID)
8580 (ReSID)
8580 + digi boost (ReSID)
USBSID-Pico
ReSID sampling method
Fast
Interpolation
Resampling
Fast resampling
Extra SIDs
1
SID I/O addresses
SID #2 $d420
SID #3 $d440
SID #4 $d460
SID #5 $de80
SID #6 $df40
SID #7 $de40
SID #8 $dfc0
Enable SID filter emulation
6581 passband 90%
6581 gain 97%
6581 bias +0,50mV
USBSID options
Enable read mode
Enable stereo mode
USBSID buffer diff
32
64
128
256
USBSID buffer size
512
1024
2048
4092
8192
16384
Save settings on exit
Confirm on exit
Pause when showing settings
Close

## x64sc SDL2

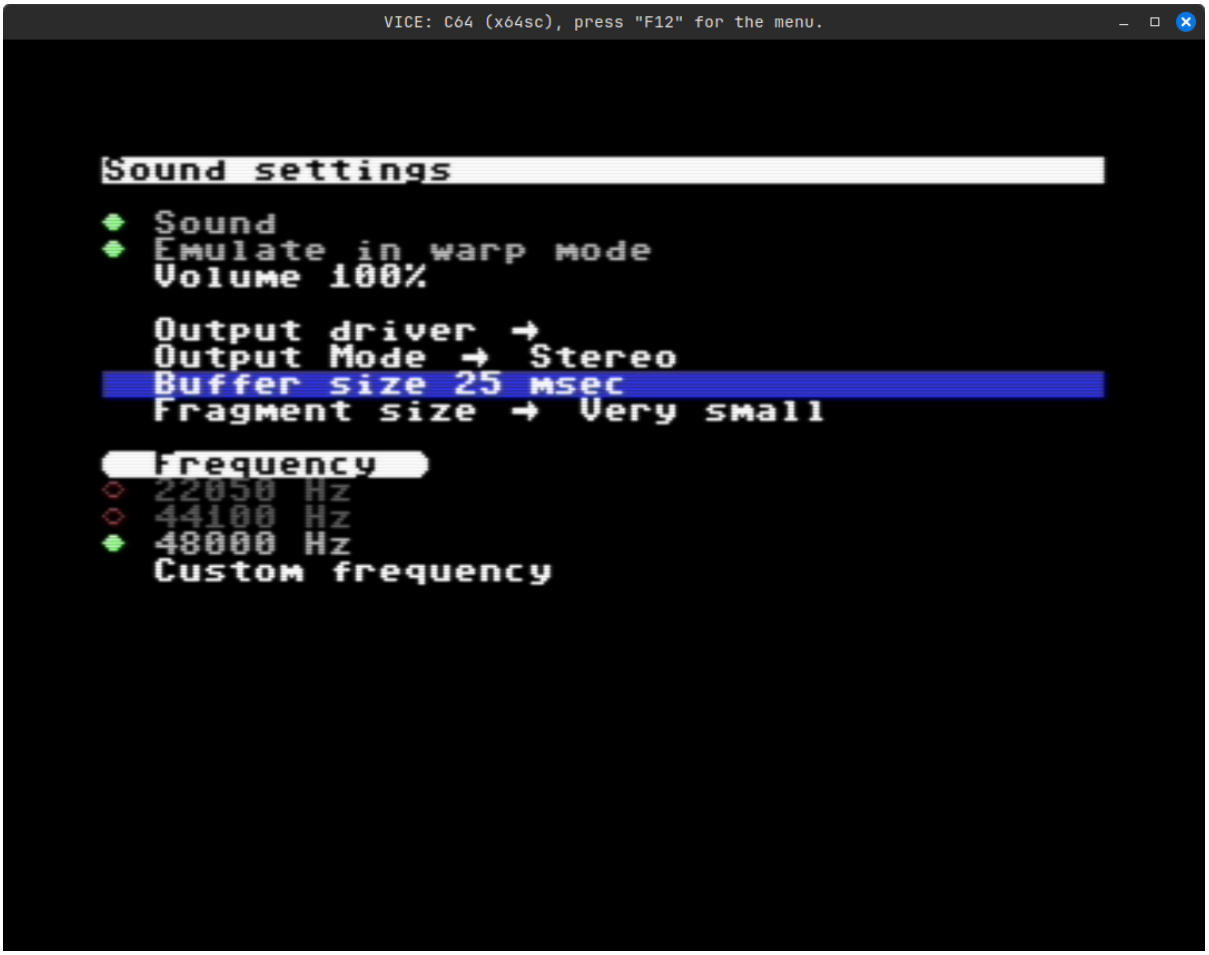
Press F12 (Linux/Windows) / F10 (MacOs) to open the settings menu.

**Sound configuration**

The sound configuration will require tuning for the computer you use it on.

In the screenshot below you can see that for this system I use the `pulse` audio driver and have the `Buffer size` set to 25 milliseconds and `Fragment size` set to very small.

These 3 settings can make or break the audio quality.




To get to these settings follow the following settings path:

`Sound settings` ⇒ `Output driver` ⇒ Select the driver you want to use

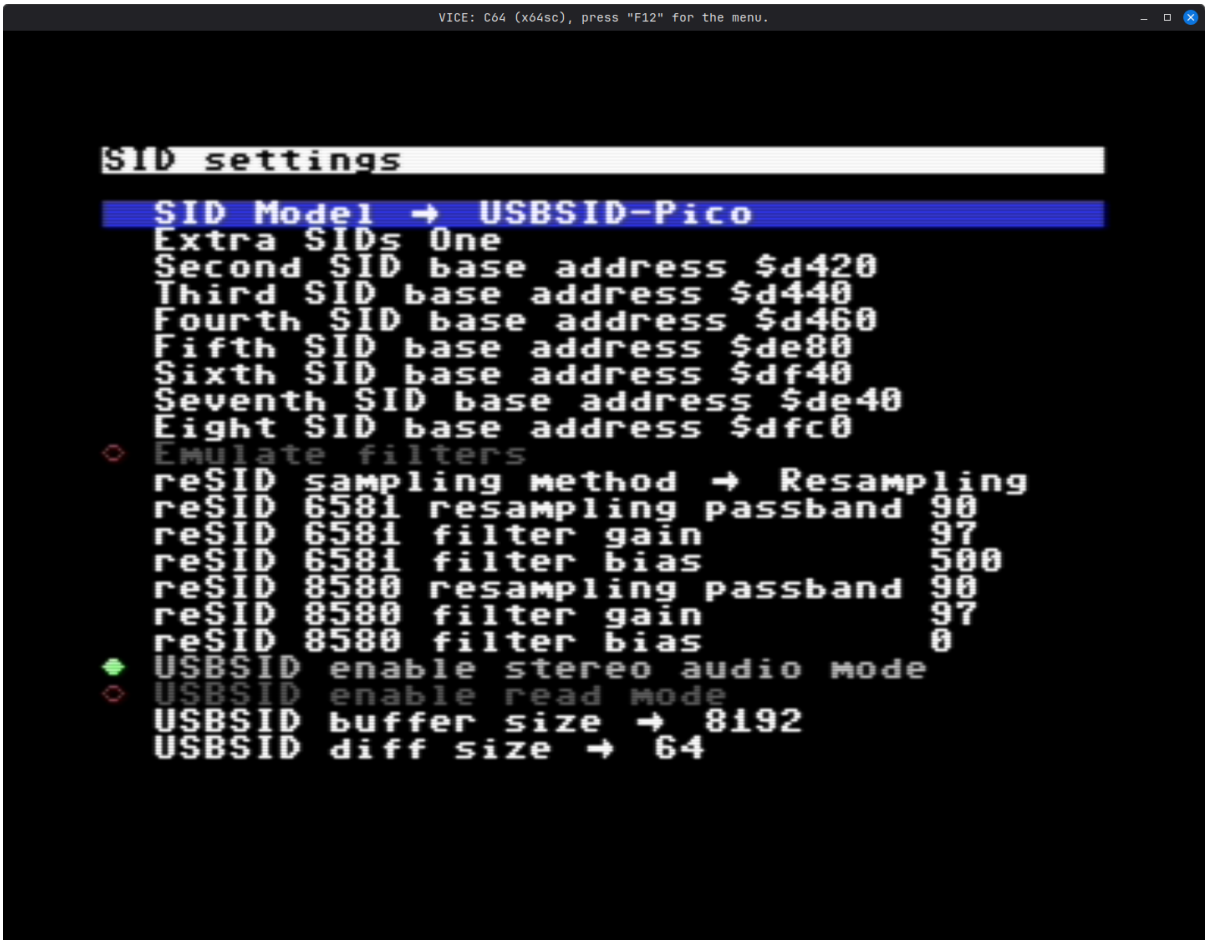
`Sound settings` ⇒ `Buffer size` ⇒ Select the milliseconds you want to use

`Sound settings` ⇒ `Fragment size` ⇒ Select the size you want to use

**SID configuration**

For any audio to come out of the board you *need* to select USBSID-Pico as SID engine.

If you want to use more the one SID then you need to select the amount of `Extra SIDs` and set their I/O addresses.




To get to these settings follow the following settings path:

`Machine settings` ⇒ `Model settings` ⇒ `SID settings` ⇒ `SID model` ⇒ `USBSID-Pico`

`Machine settings` ⇒ `Model settings` ⇒ `SID settings` ⇒ `Extra SIDs` ⇒ select the amount of extra SID's and set their addresses

## vsid GTK

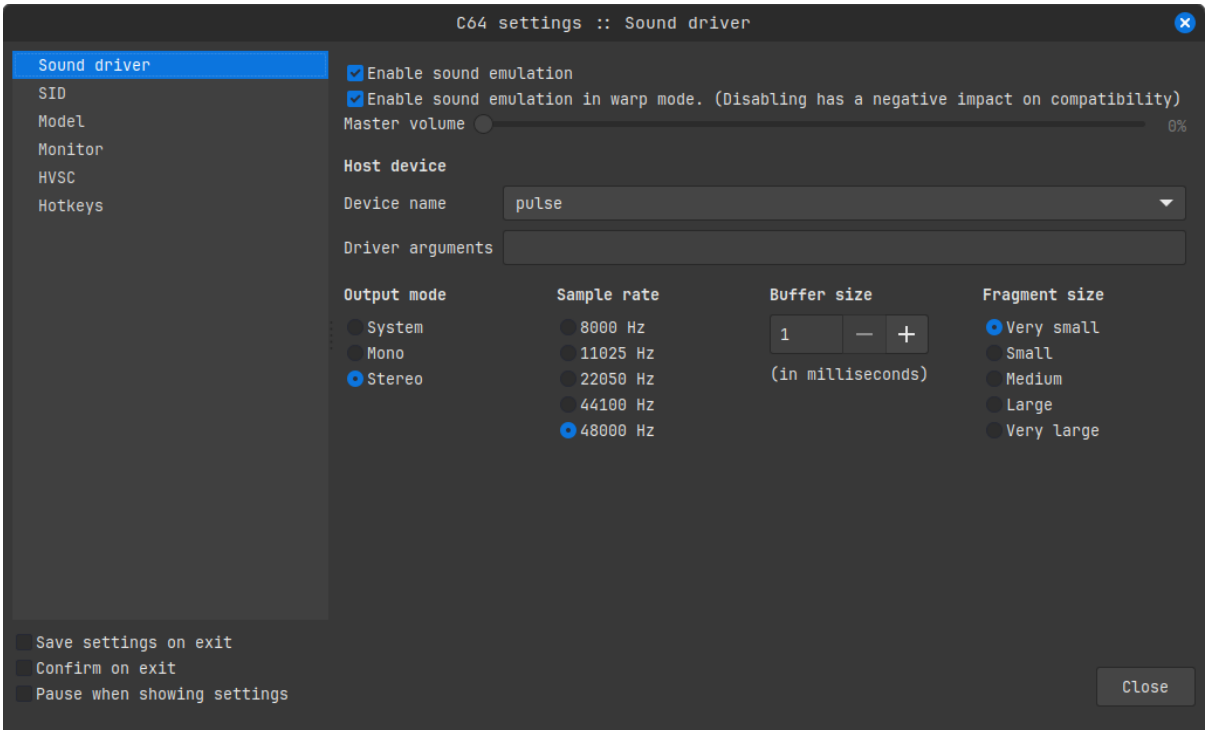
ALT+O or click Preferences/Settings to open the settings menu.

**Sound configuration**

The sound configuration will require tuning for the computer you use it on.

In the screenshot below you can see that for this system I use the `pulse` audio driver and have the `Buffer size` set to 1 millisecond and `Fragment size` set to very small.

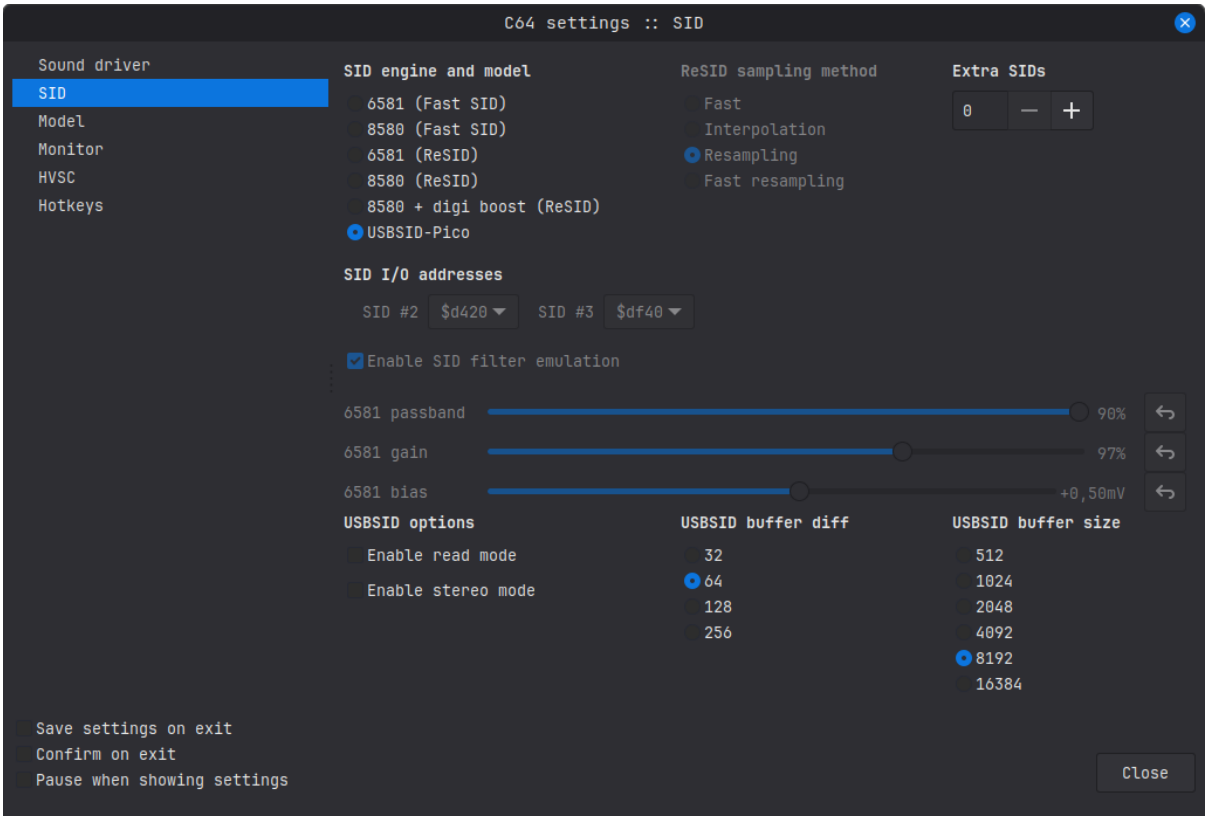
These 3 settings can make or break the audio quality.




**SID configuration**

For any audio to come out of the board you *need* to select USBSID-Pico as SID engine.

If you want to use more the one SID then you need to select the amount of `Extra SIDs` and set their I/O addresses.


C64 settings :: SID
Sound driver
SID
Model
Monitor
HVSC
Hotkeys
SID engine and model
6581 (Fast SID)
8580 (Fast SID)
6581 (ReSID)
8580 (ReSID)
8580 + digi boost (ReSID)
USBSID-Pico
ReSID sampling method
Fast
Interpolation
Resampling
Fast resampling
Extra SIDs
0
SID I/O addresses
SID #2 $d420
SID #3 $df40
Enable SID filter emulation
6581 passband 90%
6581 gain 97%
6581 bias +0,50mV
USBSID options
Enable read mode
Enable stereo mode
USBSID buffer diff
32
64
128
256
USBSID buffer size
512
1024
2048
4092
8192
16384
Save settings on exit
Confirm on exit
Pause when showing settings
Close

## vsid SDL2

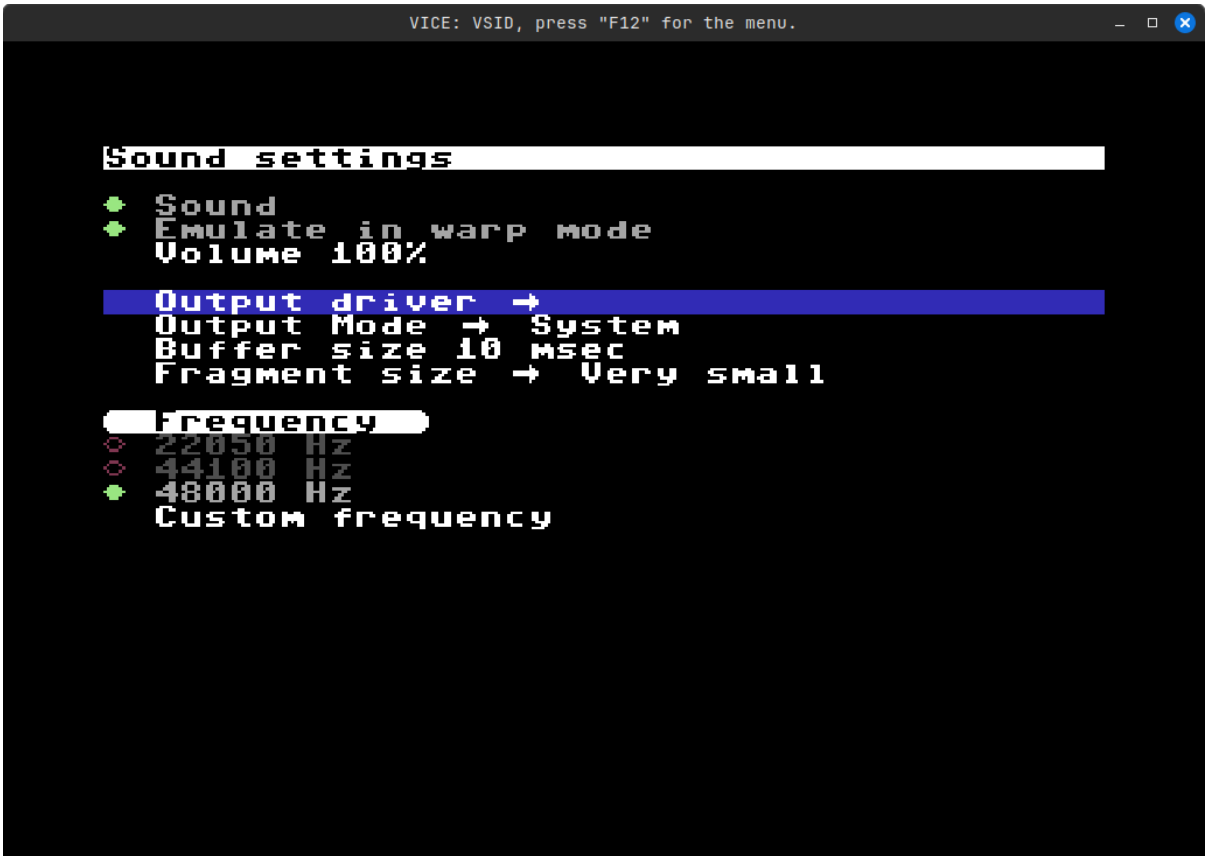
Press F12 (Linux/Windows) / F10 (MacOs) to open the settings menu.

**Sound configuration**

The sound configuration will require tuning for the computer you use it on.

In the screenshot below you can see that for this system I use the `pulse` audio driver and have the `Buffer size` set to 10 milliseconds and `Fragment size` set to very small.

These 3 settings can make or break the audio quality.




To get to these settings follow the following settings path:

`Sound settings` ⇒ `Output driver` ⇒ Select the driver you want to use

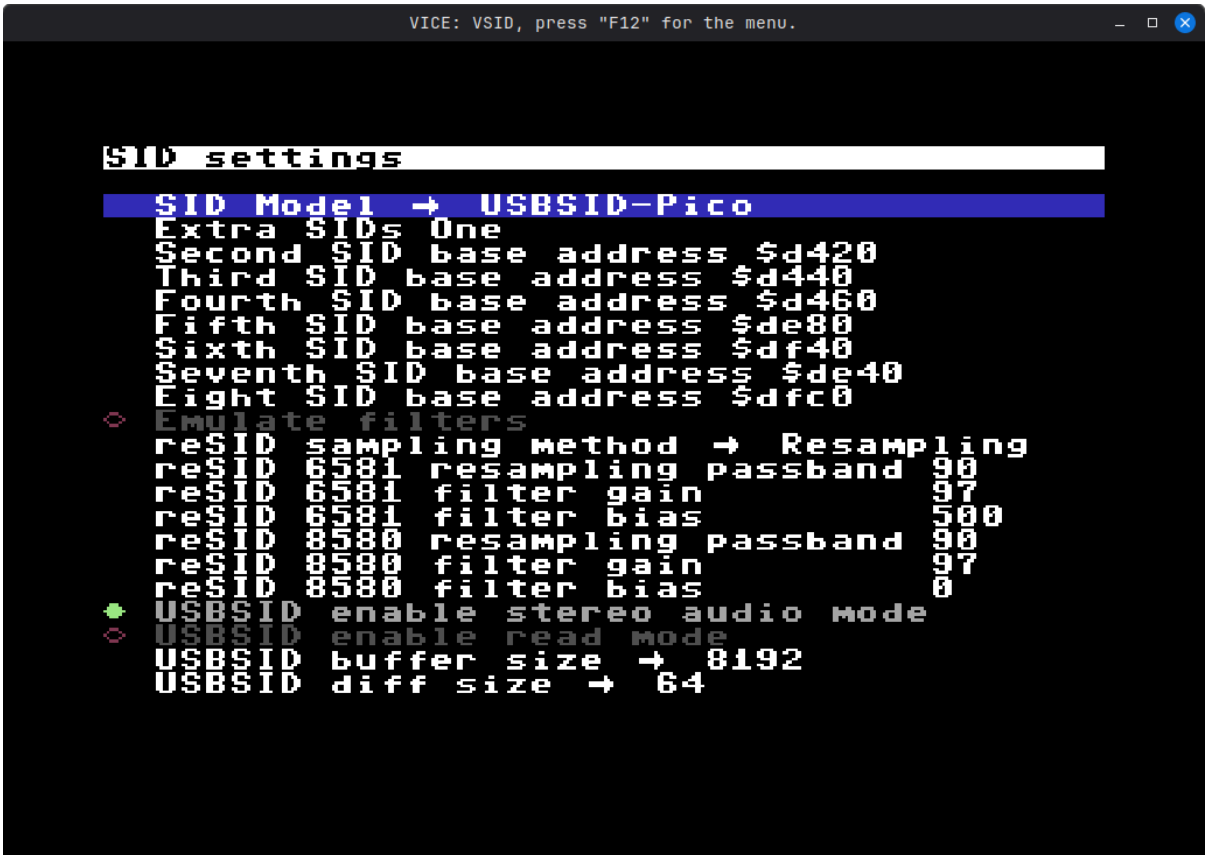
`Sound settings` ⇒ `Buffer size` ⇒ Select the milliseconds you want to use

`Sound settings` ⇒ `Fragment size` ⇒ Select the size you want to use

**SID configuration**

For any audio to come out of the board you *need* to select USBSID-Pico as SID engine.

If you want to use more the one SID then you need to select the amount of `Extra SIDs` and set their I/O addresses.



To get to these settings follow the following settings path:

`SID settings` ⇒ `SID model` ⇒ `USBSID-Pico`

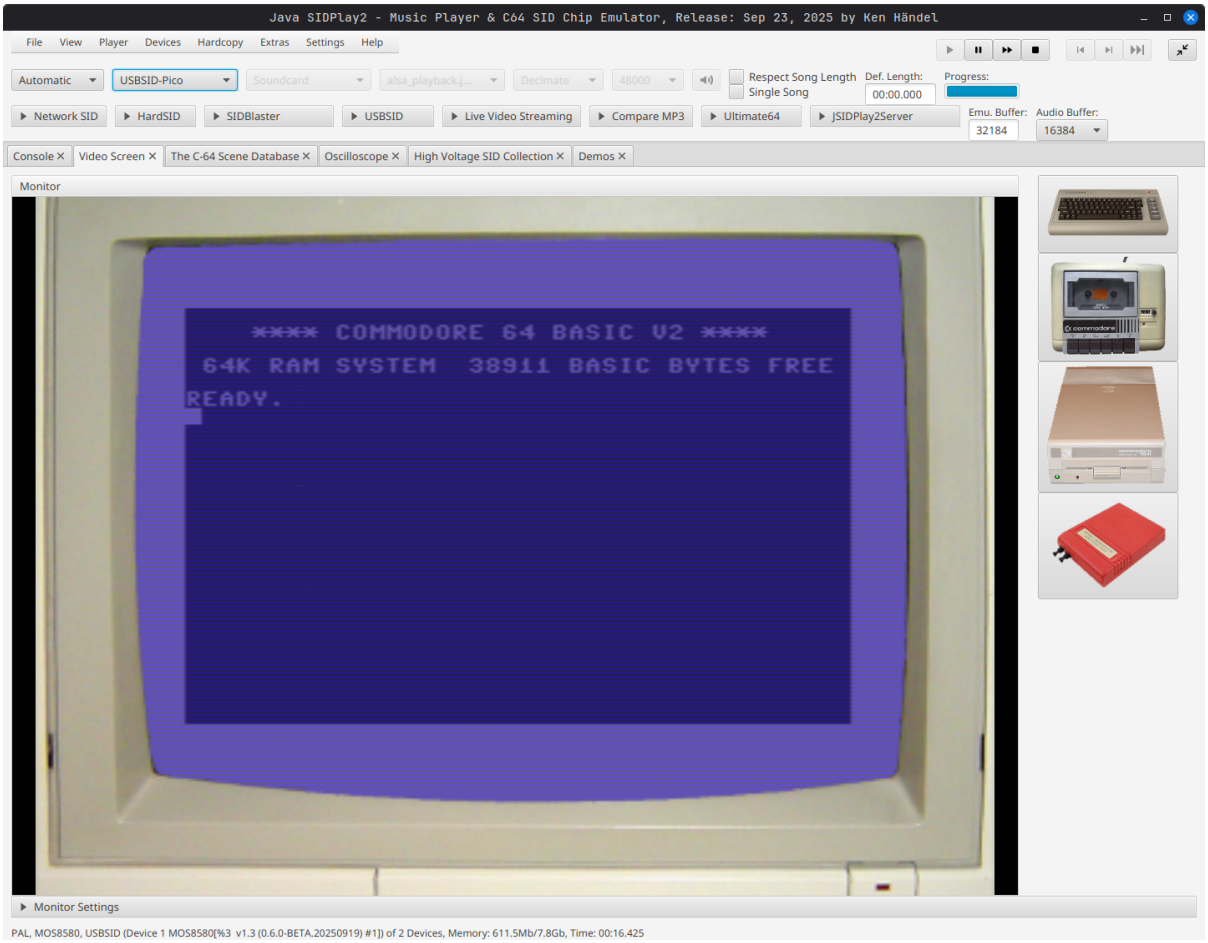
`SID settings` ⇒ `Extra SIDs` ⇒ select the amount of extra SID's and set their addresses

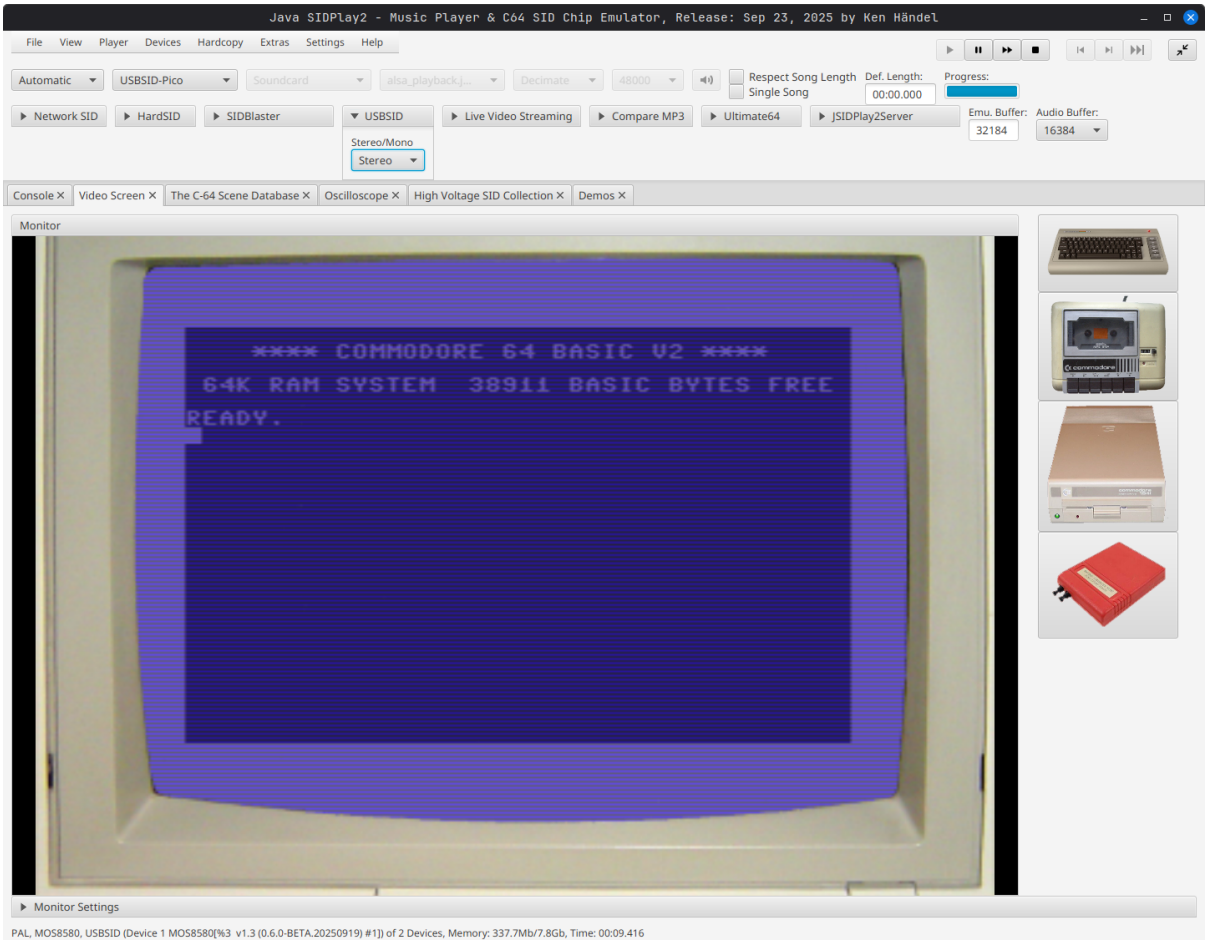
# jsidplay2

This section covers the use of USBSID-Pico with `jsidplay2`

## Settings

After starting JSidplay2 you can choose USBSID-Pico from the dropdown menu that says emulation and set the board to mono or stereo using the settings dropdown





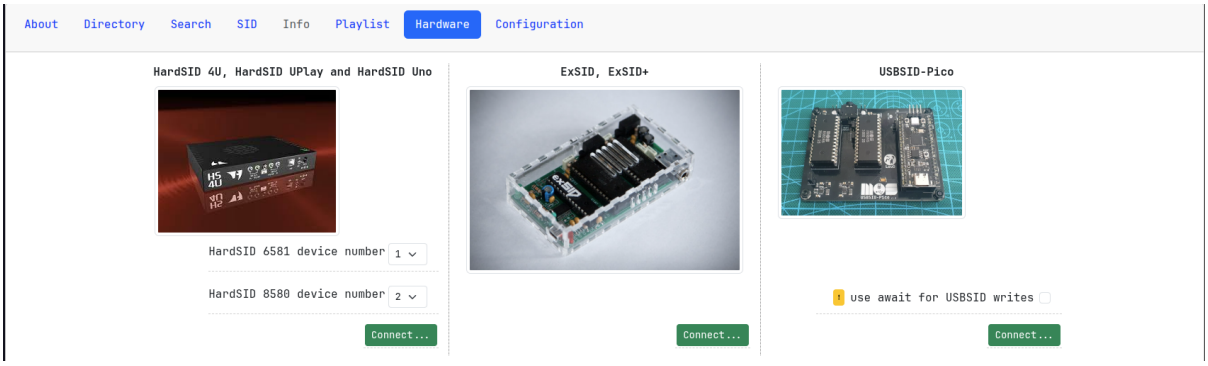

## c64jukebox

`c64jukebox` website that runs on `jsidplay2` available at haendel.ddns.net:8443/static/c64jukebox.vue

To enable USBSID-Pico in a WebUSB supporting browser (like Chrome) click on the `Hardware` tab and then click `Connect` in the USBSID-Pico box. After connecting search for any SID file in the search tab and click the play button.

**WARNING**

Because this play method is not perfected yet there will be delays when selecting a follow up tune and pressing stop or pause. This can be circumvented enabling `use await for USBSID write` on the Hardware tab, but this will severely impact playing tunes that rely on timing

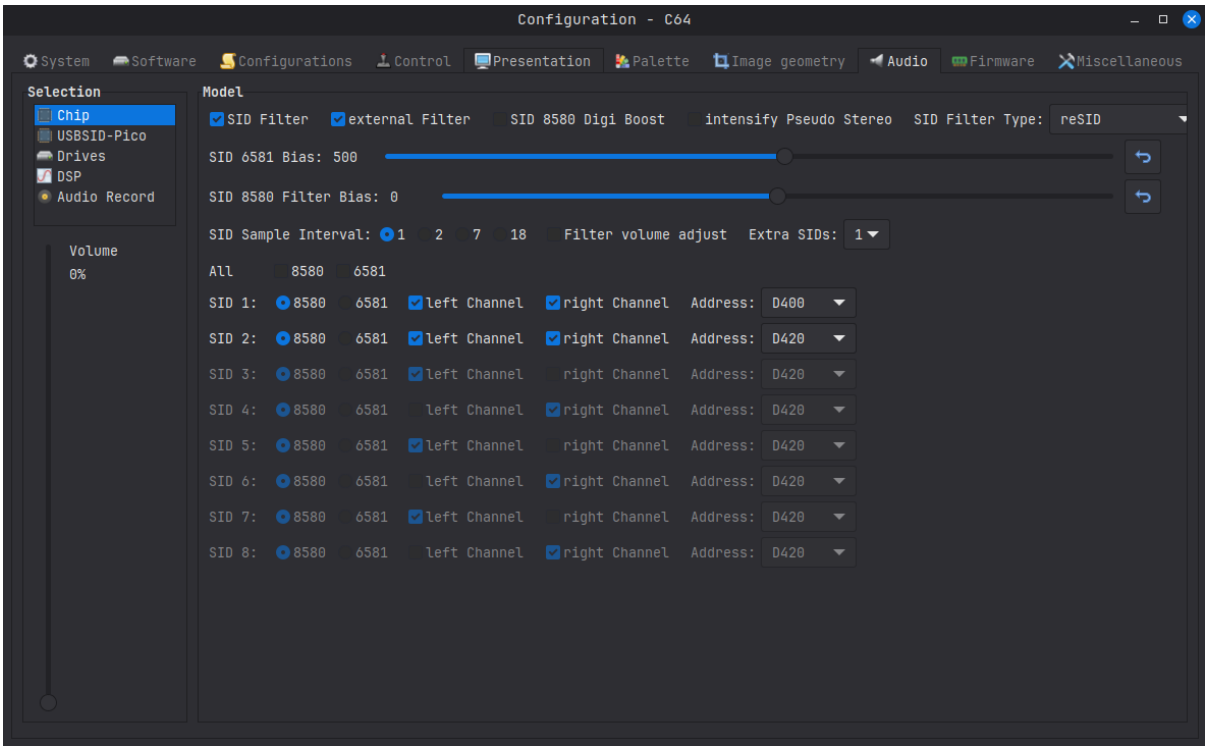


# Denise

This section covers the use of USBSID-Pico with the C64 emulator of `Denise`

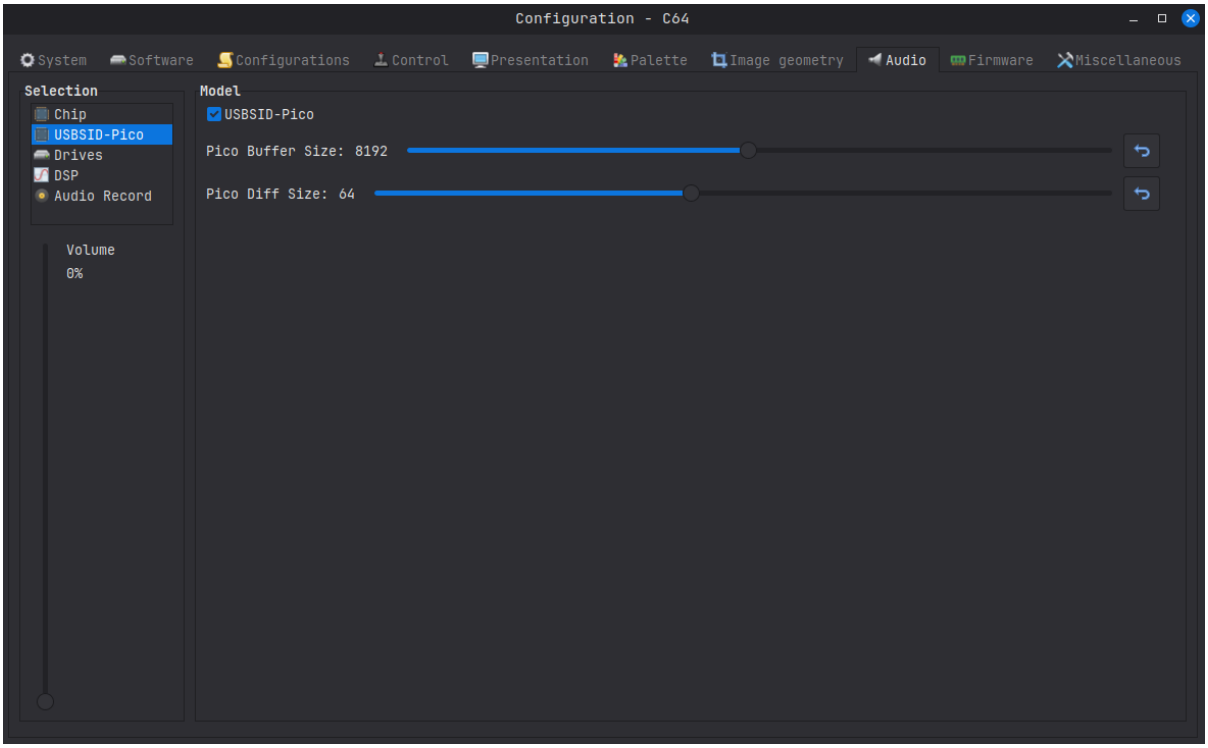
## USBSID settings

Open the C64 settings menu and go to the audio tab. Under this tab on the left side you will see an `USBSID-Pico` option
Here you can enable or disable the use of USBSID-Pico. If you have any audio issue you can try to increase the buffer and diff sizes.




## SID settings

Open the C64 settings menu and go to the audio tab. Under this tab on the left side you will see a `Chip` option
Here you can configure the amount of SID's you have on the board. Be sure to try the different settings for better output.


Configuration - C64
System
Software
Configurations
Control
Presentation
Palette
Image geometry
Audio
Firmware
Miscellaneous
Selection
Chip
USBSID-Pico
Drives
DSP
Audio Record
Volume
0%
Model
USBSID-Pico
Pico Buffer Size: 8192
Pico Diff Size: 64

# Network SID Device (NSD) clients (ACID 64, JSidplay2, SIDPlay & USBSID-Player)

`pico_w` and `pico2_w` firmware builds compiled with `_net` appear on the LAN as a Network SID Device server, the protocol ACID 64, JSidplay2, SIDPlay and USBSID-Player speak as network clients. No driver, libusb or WebUSB permission needed. Point the client at the board's IP address (or let it auto-discover the board, see below) and it plays like a USB-attached USBSID-Pico.

## Discovery

The board answers a UDP broadcast on port `6581` whose first 9 bytes are the ASCII string `SidDevice` with `SidDevice,<hostname>,<board description>`. ACID 64 and JSidplay2 use this to fill their device lists. If the board doesn't show up automatically, use its configured hostname or IP directly. The responder listens on UDP port `6581`, also when the TCP port is changed, and can be switched off with the `Discovery Enabled` setting, see the `USBSID-Pico-firmware-manual` file. The board's IP address is printed by `cfg_usbsid -wifis`.

## Connecting

Add the board as a Network SID Device host in the client's settings using its IP address and port `6581` (the default, configurable with the `cfg_usbsid` commandline tool). USB stays fully functional at the same time. Whichever transport writes to the SID bus first keeps it until it goes idle. A network client and a USB client can share the board but only one drives the SID chips at a time.

## Multi-SID

The register byte the NSD protocol sends is the USBSID-Pico SID address space (`0x00-0x1F` SID 1, `0x20-0x3F` SID 2, `0x40-0x5F` SID 3, `0x60-0x7F` SID 4), and 2SID and 4SID tunes address the correct socket without extra configuration. The protocol itself addresses up to 16 SIDs. Accesses to SID 5 and up, and to chips on a disabled socket, are ignored but keep their delay, reads of them return `0x00`.

## Write buffer

The board queues up to 4096 writes. It answers `BUSY` when about 160 ms (PAL) of cycles are queued and accepts writes again below about 120 ms. Reads, resets and clock changes wait until the queued writes are played, a SID detect read cannot overtake the writes before it.

## FM OPL

The FM OPL socket is reached with `TRY_WRITE_EX` address `$DF40` is the FM OPL address/index port and `$DF50` the data port, one entry per port write, a register write is two entries. The client enables the device with `TRY_SET_FM_OPL`, which resets the chip when it turns a disabled device on. Other `$DFxx` addresses are ignored but keep their delay. USBSID-Player v1.4.0 and older do not play FM OPL over NSD against this firmware.

### Audio-domain settings

Commands that only make sense for a software SID emulator (sample rate, stereo positioning, per-channel level, fade in/out) are accepted and acknowledged but have no effect on real hardware. The audio path (mono/stereo switch, SID socket configuration) is controlled over USB/WebUSB.

## Retrodebugger

t.b.d.

## sidplayfp

t.b.d.

# License

## Software License ~ GNUv2

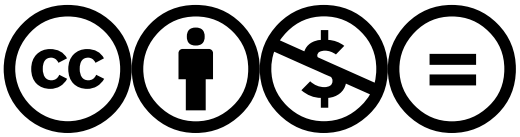
All code written by me in this repository is licensed under the terms of the GNU General Public License as published by the Free Software Foundation, version 2. Any code in this repository that is not written by me automatically falls under the licensing conditions by the authors of said code as mentioned in the source code header. For for more information about this license please read the LICENSE document in the root of the USBSID-Pico repository.

## Configtool Software License ~ EPLv2

This program and the accompanying materials are made available under the terms of the Eclipse Public License 2.0 which is available at eclipse.org.
SPDX-License-Identifier: EPL-2.0

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